

Yinxi Li (Larry)

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Education

University of Waterloo, Canada <i>M.Math. in Computer Science, Advisor: Pengyu Nie </i>	<i>May 2025 – April 2027</i>
The Chinese University of Hong Kong, Hong Kong (SAR) <i>B.Sc. in Computer Science, ELITE Stream </i>	<i>Sept 2020 – July 2024</i>
ETH Zürich, Switzerland <i>D-INFK, Exchange Student</i>	<i>Feb 2023 – Sept 2023</i>

Publications

- Yinxi Li**, Yuntian Deng, and Pengyu Nie. “TokDrift: When LLM Speaks in Subwords but Code Speaks in Grammar.” In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. [arXiv:2510.14972](#) 
- Yinxi Li**, Jiaao Chen, Fang Wu, Jiakai Yu, Heli Qi, Weihao Xuan, Haokai Zhao, Pengyu Nie, Di Jin, and Xiangru Tang. “Learning Multi-step Reasoning via Persistent Latent State Propagation.” In *Workshop on Latent & Implicit Thinking – Going Beyond CoT Reasoning*, 2026.

Project Experience

- Unit Test Transplantation with Retrieval-Augmented Generation** *Sept 2024 – Dec 2024*
- Developed a RAG pipeline for test transplantation to improve unit test generation in API frameworks
 - Analyzed model performance on 2,000+ unit test cases extracted from 7 diverse open-source repositories, identifying optimization opportunities and proposing enhancement strategies
- Optimizing Structure of Vision-Language Retrieval System** *Aug 2023 – June 2024*
ELITE Stream Final Year Project, Supervisor: [Eric Lo](#) 
- Developed and integrated a proxy model and optimized retrieval structures, achieving up to 50% faster object detection and 1000× inference speedup with minimal accuracy loss
 - Designed a vision-language retrieval system for large-scale video datasets, enabling efficient and diverse top-K results with reduced computational complexity from $\mathcal{O}(n^2d)$ to $\mathcal{O}(nd)$
 - Conducted comprehensive evaluations, including baseline comparisons and ablation studies, demonstrating superior retrieval accuracy, diversity, and architectural effectiveness
- Adaptive Streaming for Short-Video Services** *June 2022 – Aug 2022*
CUHK Summer Research, Supervisor: [Lee Yiu-Bun](#) 
- Implemented and evaluated multiple adaptive bitrate (ABR) algorithms (including BBA, BOLA, and the RL-based Pensieve), and tuned the simulator to support focused testing on short video samples

Competition

- World University Students’ Digital Intelligence Competition Invitational Tournament (by Tencent)** *April 2023 - July 2023*
- Ranked first among overseas institutions, Mentor: [Tan LEE](#) *
- Developed a multi-agent reinforcement learning framework for Honor of Kings, with optimized reward functions and customized agent strategies

Research Internship Experience

- Research Assistant** *Remote*
Southern University of Science and Technology, Supervisor: [SONG Xuan](#)  *Aug 2024 - Oct 2024*
- Developed an expert knowledge collaboration system for the automotive quality inspection value chain based on large language model and retrieval-augmented generation

Research Intern

Research Institute of Tsinghua University in Shenzhen

*Shenzhen, China
April 2024 - June 2024*

- Implemented statistical, ML, and deep learning methods for multivariate time-series anomaly detection
- Utilized a variational autoencoder for anomaly detection, improving customer load monitoring

Research Intern

Institute of Optics, Fine Mechanics and Physics, Chinese Academy of Sciences

*Changchun, China
June 2020 - Aug 2020*

- Optimized object detection of moving objects in specific directions in videos

Research Community Service

Teaching Assistant: CS 246, CS346, CS446 (25 Spring, Fall, 26 Winter, UWaterloo)

Sub-Reviews: ASE 2025, ARR Oct 2025

Association Activities Experience

CUHK Mainland Undergraduate Association (19th & 20th cabinet)

Aug 2021 - July 2023

- Principal of Information Department: Managed communication channels
- Assistant of Sports Department: Coordinated Sports Festival and e-sports tournaments

CUHK Mainland Undergraduate Association Orientation Camp

Aug 2021 - July 2023

- Group Leader(2021-23): Mentored 7 groups and supported 60+ freshmen in academic and campus life

Additional Information

Open Source Contributions: [TokDrift](#) , [Arxiv Paper Agent](#) 

Selected Coursework: Machine Learning for SE, Computational Linguistics, Mining & Analysis of SE Data, Foundations of Optimization, Machine Learning Theory, Fundamentals of AI, Rigorous Software Engineering, Introduction to ML, Data Modelling and Databases

Programming Languages: Python, Java, Shell Script